Priority-Conditioned Retention: A Controllable Middle-Layer Memory Circuit in Pretrained Mamba

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Abstract

State-space models (SSMs) such as Mamba compress an entire history into a fixed-size continuous hidden state, achieving linear-time inference at the cost of associative recall: unlike attention, they cannot losslessly retrieve an arbitrary past token. Prior work attributes this to a capacity bound but leaves open whether the practical failure is a *storage* limit or a *control* limit. We show that a pretrained Mamba-790m already contains a controllable, spatially localized memory circuit, accessible at inference with no fine-tuning. By scaling the pre-softplus input to the Δ gate at positions downstream of a target, we reduce the effective decay applied to that target and recover associative recall on a capacity-pressured multi-query associative-recall (MQAR) task ($0.53 \rightarrow 0.73$ at the optimal operating point). The effect is dose-dependent, target-specific, and bidirectional, and it is localized to a middle-layer band. Fine-grained ablation identifies a minimal 4-layer core at L19–22 that matches the 8-layer result (0.70 vs. 0.67, equal within noise at n=150) at nearly identical perplexity cost ($\times 1.08$ vs. $\times 1.07$), whereas the all-layer intervention inflates perplexity $\times 1.81$. We frame this as the continuous-state analogue of importance-aware KV-cache compression and report a boundary: a literal write-quantization analogue does *not* transfer, because the superimposed state has no per-item slots to reallocate.

1 Introduction

Mamba and related selective SSMs match Transformers on many language tasks while running in linear time, but remain markedly weaker at *associative recall* — retrieving a specific value bound to a key seen earlier in context. Jelassi et al. (2024) show this stems from a fundamental capacity constraint: a fixed state cannot losslessly copy arbitrarily long inputs. Yet a capacity bound does not, by itself, explain *behavioral* recall failure at moderate load. The failure could be (a) a **control** problem (the recurrence applies a default recency decay and does not preferentially keep query-relevant items), or (b) a **storage** problem (the state is genuinely out of bits). These have opposite implications: (a) is fixable at inference; (b) is not.

This paper provides evidence for (a) in a regime where it matters. We do *not* modify or retrain the model. Instead we ask: does a pretrained Mamba already expose a handle that lets us trade its default recency policy for a target-preserving one? We find that it does, that the handle is the Δ gate, and, most importantly, that the handle is localized to a contiguous band of middle layers. Operating only on that band recovers recall at almost no cost in language-modeling quality, the behavior one expects of a localized mechanism rather than a diffuse one.

Contributions.

- A test-time intervention on the Δ gate that controllably preserves a target item in a capacity-pressured SSM, with dose-response, specificity, and bidirectionality controls (Section 4.1).

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- Localization of this control to layers 16–23 of Mamba-790m, and a *surgical* result: the 8-layer intervention matches most of the full-model recall gain at perplexity $\times 1.07$ vs. $\times 1.81$ (Section 4.2). Fine-grained ablation further identifies a 4-layer minimal circuit (L19–22) that matches the 8-layer recall (0.70 vs. 0.67; confirmed equivalent at n=150, Experiment F) at $\times 1.08$ cost, and is both sufficient and necessary within L16–23 (Experiment G).
- A framing placing this control on a single axis with importance-aware KV-cache compression, plus a negative result: a literal write-quantization analogue fails, bounding how far the cache analogy carries (Section 5).

On scope: all positive results are for Mamba-790m on synthetic MQAR. We observe non-replication on Mamba-1.4b and fragility on natural language (Section 6).

2 Background and Framing

The Δ gate as a retention knob. In a selective SSM the state updates as $h_t = \bar{A}_t h_{t-1} + \bar{B}_t x_t$, with $\bar{A}_t = \exp(\Delta_t A)$ and $A < 0$, so $\Delta_t > 0 \Rightarrow \bar{A}_t \in (0, 1)$. The contribution of token i surviving to a query at position T is therefore $\bar{B}_i x_i \cdot \prod_{\tau > i} \bar{A}_\tau$. Two facts follow. First, the default policy is recency-biased: older contributions decay geometrically. Second, the *survival* of an item is governed by the *downstream* decays $\prod_{\tau > i} \bar{A}_\tau$, not by the item's own write alone. Reducing Δ at positions after a target pushes those $\bar{A}_\tau \to 1$, slowing the decay the target experiences — a preservation lever requiring no weight change.

Discrete $\leftrightarrow$ continuous importance allocation. The KV-cache compression literature (Liu et al., 2024; Hooper et al., 2024; He et al., 2024; Zandieh et al., 2026) implements one principle: spend more bits on more important tokens. This presumes a slotted cache with random access. Mamba has no slots — all items are superimposed in one vector. We therefore treat "importance $\to$ retained information" as architecture-agnostic and instantiate it in Mamba not as per-token bit allocation but as *differential retention* through the Δ gate. We name this framing Priority-Conditioned Retention (PCR). Section 5 reports where the analogy holds and breaks.

3 Method

Task. Multi-query associative recall (MQAR) with random single-token key/value words (no semantic priors). A prompt presents N key–value pairs followed by a query key; the model must emit the bound value. Capacity pressure is induced by increasing N; we query the first pair as the default target. As N grows the base model degrades smoothly (recall $0.77 \to 0.53 \to 0.40$ for N=16, 32, 64); we adopt N=32 (base recall 0.53) as the operating point that balances pressure and headroom.

Intervention. We register forward hooks on each layer's `dt_proj` and multiply its pre-softplus output by a per-position gain g. Setting $g < 1$ ("other-gain") at positions downstream of the target's value reduces their effective decay. A layer mask restricts the intervention to a chosen subset of layers. The model runs on its pure-PyTorch sequential path (fused kernels intentionally not installed) so the Δ pathway is hookable. No parameters are updated.

Model and metrics. `state-spaces/mamba-790m-hf` (48 layers, fp32). We report top-1 recall over 30 random seeds per condition, and perplexity on a fixed held-out passage under the same gain, relative to the unmodified model.

4 Results

4.1 Test-time control

Sweeping other-gain at N=32 over all layers yields a clear inverted-U with an optimum at g=0.8, where recall rises $0.53 \to 0.73$, then collapses for $g \leq 0.6$ as integration halts (Figure 1). The non-monotonicity is mechanistically expected: lowering Δ both slows decay ($\bar{A}_\tau \to 1$, helpful) and shrinks the write $\bar{B}_\tau = \Delta_\tau B_\tau$ (harmful); past $g \approx 0.7$ the latter dominates and integration stalls.

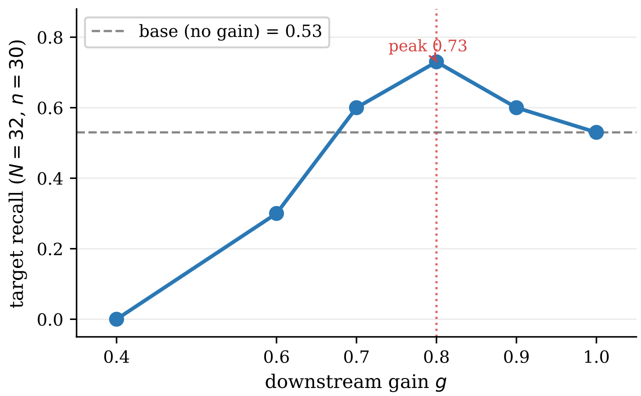

Figure 1: Dose–response of the Δ-gate intervention (Mamba-790m, MQAR N=32). Recall peaks at g=0.8 (0.53 → 0.73) and collapses when Δ is over-suppressed, confirming Δ as a graded retention knob.

Table 1: Specificity and direction controls at N=32, g=0.8 (base recall 0.53). Only targeted downstream suppression helps.

Condition	Target recall
Base (no intervention)	0.53
Downstream suppression (g=0.8)	**0.73**
Random-position suppression (g=0.8)	0.50
Upstream / increase Δ	<0.53

The effect is **target-specific**: applying g=0.8 to downstream positions yields 0.73, whereas the same gain budget on *random* positions yields only 0.50 (near base), ruling out a generic "more gain helps" artifact (Table 1). It is also **bidirectional**: *increasing* Δ degrades recall below base, so the sign of the intervention maps to the sign of the effect.

4.2 Layer localization and the surgical circuit

Restricting g=0.8 to 8-layer bands isolates the responsible depth: recovery concentrates in **L16–23** (+0.13), with a secondary contribution at L24–31 (+0.07) and nothing elsewhere (Figure 2). Crucially, the localization is also *cheaper*: Table 2 shows the 8-layer intervention captures most of the recall gain at near-zero language-modeling cost, while the all-layer version pays a large perplexity tax. A targeted edit to the memory circuit is almost free while a global one is not, which is what we would expect from a dedicated mechanism rather than a diffuse property.

To verify that L16–23 is not an artifact of the N=32 operating point, we swept the band localization at $N \in \{16, 32, 64\}$ (Experiment A). L16–23 is the top 8-layer band at every pressure level, and its advantage grows with load (Table 2). At N=64 a notable reversal emerges: the surgical circuit (0.60) overtakes the all-layer intervention (0.55). Per-band inspection shows why: L8–15 and L32–39, neutral at low N, shift to −0.20 and −0.25 at N=64. Off-target harm grows with capacity pressure and the surgical variant avoids it entirely. Not only does L16–23 suffice; the surrounding layers actively interfere under high load.

To identify the minimal effective sub-circuit, we performed a fine-grained ablation within L16–23 (Experiment B, N=32, g=0.8). Sliding 4-layer windows across L16–27 show the recall gain is concentrated in the rear of the band: L19–22 achieves recall 0.70 (Δ+0.17), matching the full 8-layer band within sampling noise. Earlier windows are weaker (L16–19: Δ+0.10; L17–20, L18–21: Δ+0.13), and L23–26 is actively harmful (−0.10). Individual-layer inspection confirms L23 as a boundary layer (−0.13 alone). The 4-layer sub-band L19–22 therefore constitutes a *minimal effective circuit*: half the layers, nearly identical PPL cost (×1.08 vs. ×1.07), and recall statistically equivalent to the 8-layer band (Table 3, Experiment F). The surrounding layers L16–18 and L23 dilute rather than contribute. Experiment G further confirms that L19–22 is not merely sufficient

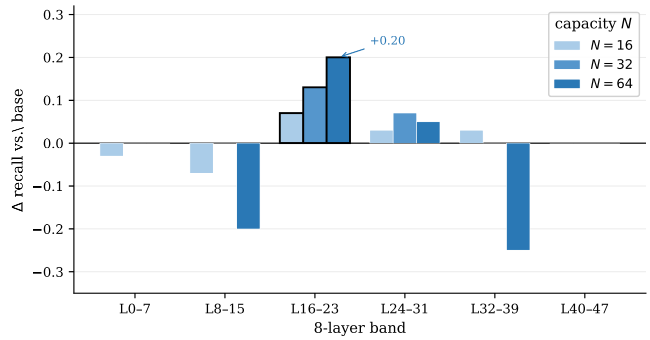

Figure 2: Layer-band localization across capacity levels (Experiment A). Applying g=0.8 to one 8-layer band at a time, L16–23 is the top band at every N. At N=64, off-target bands L8–15 (-0.20) and L32–39 (-0.25) turn actively harmful, making the surgical choice more important under pressure.

Table 2: Test-time Δ-gate intervention across capacity levels (Mamba-790m, g=0.8, base ppl $\approx$ 232; n=30 for $N \leq 32$, n=20 at N=64). PPL cost is N-independent. L16–23 holds a positive advantage at every level and overtakes the all-layer intervention at N=64 (**0.60** $>$ 0.55) as off-target layers turn harmful; the 4-layer minimal core L19–22 (Section 4.2) is strongest at N=32 but regresses below base at N=64. The knockout row tests necessity (L19–22 removed from L16–23).

Intervention	N=16	N=32	N=64	PPL
Base (no gain)	0.77	0.53	0.40	×1.00
All layers, g=0.8	0.90	0.73	0.55	×1.81
L16–23 (8L), g=0.8	0.83	0.67	**0.60**	×**1.07**
L19–22 (4L), g=0.8	0.87	0.70	0.35	×1.08
Necessity (knockout, N=32):				
$\{16, 17, 18, 23\}$ (L19–22 removed)	—	0.43	—	—

but *necessary*: removing it from L16–23 (leaving only layers $\{16, 17, 18, 23\}$) collapses the gain to $\Delta{-0.10}$ below base (Table 2, knockout row).

Robustness and statistical validation (Experiments F–H). We re-ran the load-bearing N=32 comparisons at n=150 (Table 3): both circuits lift recall well above base (L16–23 0.780, L19–22 0.773), and their 95% Wilson intervals overlap heavily, so the 0.03 gap seen at n=30 is sampling noise; we characterise L19–22 as *matching* L16–23, not exceeding it. (The n=150 base of 0.620 likewise places the n=30 estimate of 0.53 on the low side of its sampling distribution, leaving the intervention deltas intact.) Across capacity levels the two circuits diverge (Table 2, Figure 2): the 4-layer core matches the 8-layer band at $N{\leq}32$ but *regresses below base* at N=64 ($0.35 < 0.40$), whereas L16–23 stays positive and reaches its largest advantage there (+0.20; a dedicated $N \in \{8, \ldots, 64\}$ sweep (Experiment H) confirms the protected-vs-base gap peaks at N=64, +50% relative). L19–22 is thus the efficient choice at the operating point, L16–23 the robust choice under sustained pressure.

Scale check: Mamba-130m (Experiment C). To assess whether a proportional circuit exists in a smaller model, we applied the same 4-layer window sweep to Mamba-130m (24 layers, N=32, g=0.8). Under the proportional-depth hypothesis, the 790m circuit at L19–22 (≈40–47% depth) would predict a counterpart near L9–11 in 130m. The result (Table 4) does not support this: the best 4-layer window is L13–16 (57–70% depth), shifted well downstream of the prediction, and the peak gain is Δ+0.06, less than half the 790m effect. Eleven of 21 windows are actively harmful, and the all-layer intervention degrades both recall (0.43, below base 0.57) and language quality (×1.31). Notably, L13–16 reduces perplexity (×0.95), suggesting that Δ is already elevated by default in those layers of 130m. Fine-grained ablation within L13–16 finds a weakly superior 2-layer sub-window (L14–15, Δ+0.03), with individual layers each contributing nothing above base

Table 3: Statistical validation at N=32, g=0.8 (n=150, 95% Wilson CI). The 0.03 gap between L19–22 and L16–23 at n=30 falls within noise; both circuits are equivalent. (Experiment F)

Condition	Hits/n	Mean	95% CI
Base (no gain)	93/150	0.620	$[0.54, 0.69]$
L16–23 (8L)	117/150	0.780	$[0.71, 0.84]$
L19–22 (4L)	116/150	0.773	$[0.70, 0.83]$

Table 4: Scale comparison at N=32, g=0.8. The 790m minimal circuit does not transfer proportionally to 130m: the optimal window shifts deeper, the recall gain is substantially weaker, and PPL behavior reverses.

Model	Layers	Best circuit	Depth	Δ recall	PPL
Mamba-790m	48	L19–22 (4L)	40–47%	+0.17	×1.08
Mamba-130m	24	L13–16 (4L)	57–70%	+0.06	×0.95

(Δ+0.00), indicating no single-layer concentration. The proportional-depth prediction fails, and the effect magnitude is substantially reduced. We treat these results as a boundary condition rather than a positive replication (Section 6).

JRT baseline (Experiment D). Just Read Twice (JRT; Arora et al. 2024) is a natural training-free baseline: repeat the key–value context before the query so the model compresses it a second time, at a cost of $\approx \times 2$ sequence length. Table 5 compares JRT, PCR (L19–22, g=0.8), and their combination across $N \in \{16, 32, 64\}$.

JRT is a substantially stronger recall lever than PCR. At N=16 and N=32 it achieves perfect recall (1.00), while PCR peaks at 0.87 and 0.70 respectively. At N=64, JRT retains a large advantage (0.95), whereas the L19–22 circuit *regresses below base* ($0.35 < 0.40$): the minimal 4-layer circuit is not robust under sustained capacity pressure. (The 8-layer band L16–23 fares better at N=64; see Table 2.) Combining JRT and PCR yields no benefit beyond JRT alone — gains are non-additive (Δ+0.47 combined vs. sum Δ+0.64) because JRT already saturates recall at $N \leq 32$.

PCR's distinct advantage is zero sequence overhead (×1.00 vs. ×1.97 for JRT at N=32). The comparison therefore identifies a clear operating-point split: JRT is preferred when doubled context is affordable; PCR is preferred when sequence budget is fixed. At N=64, neither is satisfactory, and a combined JRT+PCR does not recover the shortfall.

De-confounded graded priority (Experiment E). The exploratory graded result (Section 4.2) applied per-write-position gains while always querying the earliest pair, confounding other-gain level with write position. Experiment E separates these factors: each trial randomly assigns P1/P2/P3 priority labels to three pairs, then applies downstream suppression at the assigned other-gain level, marginalizing over write position.

After de-confounding (Table 6), the monotone holds: P1 (g=0.7) achieves recall 0.13 vs. P3 (g=1.0) at 0.00, a lift of +0.10 above the no-gain random-pair baseline (0.03), while P3 shows no lift (+0.00). The zero-suppression P3 condition exactly matches the no-gain baseline, providing a clean causal signal: the other-gain level is the active lever, not position. A uniform-g control (all three levels at g=0.8) lifts all conditions above baseline but yields a weaker gradient (0.10/0.07/0.03), consistent with statistical noise at n=30 rather than a systematic position artifact. By contrast, the original confounded result (P1= 0.53) is almost entirely explained by pair 0's early write position (base recall 0.53 for pair 0 without any intervention); the differential other-gain contributes minimally to that gap. Graded allocation is therefore causally confirmed but modest in magnitude, consistent with the SSM shared-decay constraint: all pairs traverse the same decay trajectory, and downstream suppression shifts one pair's decay floor at the cost of not shifting others.

Table 5: JRT vs. PCR recall across capacity levels (g=0.8, L19–22 for PCR; n=20 at N=64, n=30 otherwise). Sequence overhead is measured at N=32 (base = 66 tokens, JRT = 130 tokens). JRT saturates recall at $N \leq 32$; combining JRT+PCR adds nothing. At N=64, PCR alone regresses below base.

Method	N=16	N=32	N=64	Seq. overhead
Base	0.77	0.53	0.40	×1.00
JRT (Arora et al., 2024)	**1.00**	**1.00**	**0.95**	×1.97
PCR (L19–22), g=0.8	0.87	0.70	0.35	×**1.00**
JRT + PCR	1.00	1.00	0.95	×1.97

Table 6: De-confounded graded priority (N=32, n=30, L19–22). Priority labels are randomly assigned to pairs each trial; downstream suppression is applied per priority level. The zero-suppression P3 (g=1.0) matches the no-gain baseline, confirming other-gain as the causal lever. The original confounded result is dominated by write-position, not other-gain.

Condition	P1	P2	P3	Monotone
Random-pair base (no gain)	0.03	0.03	0.00	—
Uniform g=0.8 (all same)	0.10	0.07	0.03	✓
De-confounded (g=0.7/0.85/1.0)	**0.13**	**0.07**	**0.00**	✓
Original confounded (cell 12)	0.53	0.17	0.17	—

5 The Discrete–Continuous Boundary

If retention is one lever, *precision* (the literal KV-quant analogue) is a candidate second lever: quantize the write of low-priority tokens to fewer bits. We tested this by quantizing the per-position convolution output (the token's written representation) to k bits. Single-item precision is not a graceful knob: target recall is flat from 16 $\rightarrow$ 8 bits, then cliffs (8-bit 0.53 $\rightarrow$ 4-bit 0.03 $\rightarrow$ 0). Reallocation fails outright: quantizing *distractor* positions to 4 bits drives target recall to 0.00 (Figure 3, Table 7): it corrupts the shared computation rather than freeing capacity.

The negative result is still informative. Because the state is a superposition with no random-access slots, lowering one item's write precision either does nothing ($\geq$ 8-bit) or injects noise the whole sequence shares ($\leq$ 4-bit). There is no per-item bit budget to reallocate. The importance-allocation principle thus transfers to Mamba *only* through retention (Section 4.2), not through quantization. This is a concrete boundary on the KV-cache analogy and an empirical instance of the "no slots" property that distinguishes continuous-state memory from a discrete cache.

6 Limitations and Scope

Single model: all positive results are for Mamba-790m. Scale checks in both directions fail to replicate the circuit. *Larger (1.4b):* all layer-band $\Delta \leq 0$ despite ample headroom — no positive circuit at a comparable pressure regime. *Smaller (130m):* a weak effect exists at L13–16 (Δ+0.06, vs. +0.17 for 790m) but at an anomalous relative depth (57–70% vs. 40–47%), with PPL decreasing rather than increasing (×0.95), suggesting different layer dynamics at this scale. The L16–23 circuit appears to be a property specific to the 790m scale; we do not claim a universal depth or mechanism. **Synthetic task:** effects are robust on random-token MQAR but fragile on natural-language multi-fact recall, where the intervention can hurt. **Graded priority is real but weak:** a de-confounded experiment (Experiment E) confirms that other-gain level causally controls retention independent of write position — P1 (g=0.7) achieves 0.13 vs. P3 (g=1.0) at 0.00 after marginalizing over pair positions (Table 6). However, the original strong confounded result (P1= 0.53) was largely a write-position artifact, and the de-confounded effect is modest in absolute terms. The SSM shared-decay constraint ($\Delta > 0 \Rightarrow \bar{A} < 1$, same trajectory for all pairs) limits item-selective allocation, consistent with Section 5; we do not claim robust multi-level allocation.

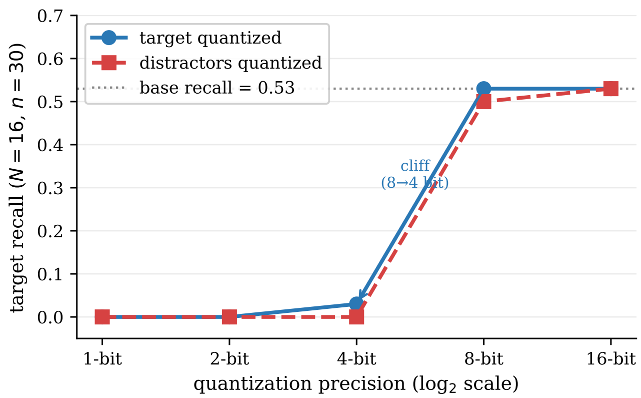

Figure 3: Write-quantization is not a graceful lever. Lowering a single item's write precision either does nothing (≥ 8 bits) or collapses recall (≤ 4 bits); quantizing distractors to free capacity instead destroys the target, evidencing the absence of per-item slots.

Table 7: Write-precision quantization at N=16. This probe uses an independent self-contained MQAR harness whose full-precision base recall is 0.53 (cf. 0.77 in Table 2); we read the *shape* of the precision curve, not absolute values. Neither targeting nor reallocation behaves like bit allocation in a slotted cache.

Quantize	16-bit	8-bit	4-bit	2-bit	1-bit
target	0.53	0.53	0.03	0.00	0.00
distractors	0.53	0.50	0.00	0.00	0.00

7 Related Work

SSM capacity. Jelassi et al. (2024) establish the copying/capacity limit; work on primacy/recency documents the default decay policy. We add a controllability result on top of the capacity picture.

KV-cache compression. KIVI, KVQuant, ZipCache and rotation-based quantizers (Zandieh et al., 2026) realize importance-aware bit allocation on slotted caches; we ask whether the principle survives the move to a slotless continuous state, and find it survives as retention but not as quantization.

Recall and the "what to keep" frontier. Prompting methods such as Just Read Twice (Arora et al., 2024) and architectures such as Based, DeltaNet and Titans (Behrouz et al., 2024) target the same problem, mostly by training new architectures or memory modules; ours is a training-free, frozen-base alternative. We directly compare against JRT (Experiment D, Table 5): JRT achieves higher recall (+0.47 at N=32) at $\times$1.97 sequence overhead; PCR provides a zero-overhead alternative (+0.17, $\times$1.00), identifying a recall-vs-cost operating-point split.

Mechanistic interpretability. Localizing a behavior to a contiguous layer band parallels circuit-style findings in Transformers; ours is, to our knowledge, an early such localization of a *controllable* memory mechanism in a pretrained SSM.

8 Conclusion and Future Work

A pretrained Mamba's associative recall is, in part, a controllable property of a specific middle-layer band, not a fixed ceiling. A surgical, training-free intervention on layers 16–23 — with a 4-layer minimal core at L19–22 that matches the 8-layer recall (0.70 vs. 0.67; confirmed equivalent at n=150) and is both sufficient and necessary (Experiment G) — recovers recall at negligible language-modeling cost. The importance-aware compression principle transfers to continuous state through retention but not through quantization. Our proposed next step is to remove the oracle: a small *question-conditioned, auto-localizing* controller trained on a frozen base, evaluated against

Belady-MIN retention efficiency under streaming queries, toward a training-free improvement of the recall–capacity frontier of pretrained SSMs.

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